Covid-19, Help-Seeking Behavior and Protection of Intimate Partner Violence Victims

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Abstract:

Although there are now several papers in the literature documenting the impact of the COVID-19 pandemic on domestic violence, this study adresses two key and previously unexplored questions about the effect of the quarantine policies on domestic violence reporting: 1) the role of third party vs victims' reporting of intimate partner violence (IPV) through hotline calls; 2) the extent to which reported domestic violence translated into protection given to victims. To address the first question, we focus on the effect of the introduction of the first lockdown in Spain on the number of reported cases of domestic violence by victims and third parties. In order to do so, we exploit province-level and monthly data on the number of calls to the dedicated emergency hotline for IPV victims between 2013 and 2020. We find that the introduction of the first lockdown was responsible for an average increase in total calls of 48%, which represents an additional 26,000 calls compared to previous years. Importantly, we also show that help-seeking behavior is at the origin of 70% of all reported calls. With respect to the second question, we provide evidence of a relative decrease of reporting to the police compared to the increase in violence that we document, which led a drop in the level of protection of victims. Despite the large increase in reported gender violence during the pandemic, we estimate that the introduction of lockdown caused a loss of at least 1,500 protection orders to victims that would, in the absence of the pandemic, have received it.

Keywords: intimate partner violence, COVID-19, help-seeking behavior

JEL: J12, J16, J18, I12,

1 Introduction

Gender-based violence – or male-based violence (MBV) - is a major public health problem as well as a violation of women's human rights. According to World Health Organization's estimates, thirty percent of women[1] have suffered from physical or sexual abuse from men at some point in their life, the majority being either their current or former partner. Intimate partner violence (IPV) is also a largely unreported event, which makes the study of the determinants of reporting and its relationship with IPV prevalence a key research question.

This already preoccupying situation has been the object of increased concern since the COVID-19 pandemic outbreak for two main reasons. First of all, the consequences of the pandemic have increased women's exposure to known risk factors of abuse such as changes in economic opportunities, (Anderberg *et al.*, 2016) stress and emotional cues (Card and Dahl, 2011)– fear of catching the disease, of losing loved ones… and the unprecedented lockdown situations imposed to counter the spread of the virus. Second of all, traditional channels of domestic abuse reporting have been deeply affected by quarantine policies: emergency shelters for victims of IPV were closed or imposed limited capacity; health centers were congested; and while police stations remained open, going to the police to seek help might have posed an even greater challenge for victims, while being on the watch of their violent partner.

In this paper we seek to contribute to the understanding of the effect of mobility restriction policies on domestic violence reporting and incidence (Peterman *et al.*, 2020; Arenas-Arroyo, Fernandez-Kranz and Nollenberger, 2020; Beland *et al.*, 2020; Campedelli, Aziani and Favarin, 2020; Kumar and Nayar, 2020; Leslie and Wilson, 2020; Piquero *et al.*, 2020;Miller, Segal and Spencer, 2020; Perez-Vincent *et al.*, 2020; Ravindran and Shah, 2020, Berniell and Facchini, 2020 to quote a few)[2]. The majority of the existing studies on this matter have found a significant increase in *reported* domestic violence during lockdown through emergency hotlines (Leslie and Wilson, 2020; Miller, Segal and Spencer, 2020; Perez-Vincent *et al.*, 2020; Ravindran and Shah, 2020) with disparities of estimated results depending on the case studied - cities vs. country - , the type of data used - hotline data vs. police data- and the time frame considered - mostly short term.

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[1] https://www.who.int/news-room/fact-sheets/detail/violence-against-women last accessed 06/07/2021

[2] See also the list updated by Dr Amber Peterman for a more exhaustive list of papers investigating the number of studies measuring the effect of COVID-19 on violence against women and children, accessible here: https://drive.google.com/file/d/1w28pmDt-lOIcI9cvaUdAMVVDzft0iI1-/view

In our paper we adress two key questions which have not yet (or only partially) been adressed in the current literature. First, the role of third party reporting in the documented increase in calls to emergency hotlines in a context where neighbors were more likely to become aware of domestic violence while being at home. Addressing this question is key to properly interpret the changes in IPV reporting: does the documented increase in measured IPV steam from a mere reporting effect by third parties who were more likely to witness pre-existing situations of abuse while being at home? Did IPV victims change their reporting strategy in the context of the quarantine? To our knowledge, there is only one paper in the literature addressing this question by Ivandic *et al.* (Ivandic, Kirchmaier and Linton, 2021). Their results show that the increase in police calls on domestic violence grounds following the lockdown were entirely driven by third party calls, rather than victims. Our second main contribution lies on the use of a stringency measure of confinement policies at the local level, which we construct using mobility data from Spanish phone companies. We thereby measure differences in effective mobility reductions in a more precise way compared to the previously used google mobility indexes (Berniell and Facchini, 2020) or Oxford Coronavirus Government Response Tracker (OxCGRT). Finally, we also contribute to the understanding of the relationship between different channels of detection of IPV victims and, more specifically, on the potential substitution between reporting through calls, and reporting to the police. As reporting to the police is a necessary step for judicial protection, understanding the importance taken by reporting through calls in the overall detection system of IPV victims is crucial from a public policy point of view to assess the level of protection that victims have at each point of time.

We focus on Spain, a country which was intensively hit by both the pandemic and mobility restriction policies and study the effect of the introduction of lockdown on male-based violence against women. The object of our study goes beyond the violence committed in the domestic environment *stricto sensus* -in the same house or by the current partner. However, as male-based violence is largely committed by intimate partners (whether current of former) in the rest of the paper we will use male-based violence (MBV), gender-based violence (GBV), intimate partner violence (IPV) and domestic violence (DV) interchangeably.[3]

[3] Domestic violence is often used to refer to male-based violence or intimate partner violence, including violence which might be done by a former partner no longer living in the same domestic environment. We also use this term and include violence against women exerted outside of the *stricto sensu* domestic environment (in the same house / family).

The first lockdown was implemented on March 15th in all regions under very strict conditions and was relaxed progressively from early June. After a rather relaxed summer with milder restrictions, the second curfew was established on the 20th October as a response to the subsequent waves of the pandemic, and was not relaxed until April 2021. We explore the effect of the introduction of the first lockdown on the reported cases of MBV by victims and third parties by exploiting province-level (there are 52 provinces in Spain)[4] and monthly data on the number of calls to the dedicated emergency hotline for IPV victims between 2013 and 2020. Using a difference-in-differences strategy (with 2020 being our treated year, and March-December being our "Post" period for every year), we find that the introduction of the first lockdown was responsible for an average increase in total calls by 48%, which represents an additional 26,000 calls compared to previous years. Contrary to previous findings, we find that the implementation of the lockdown caused an increase in both third-party reporting and victims' reporting, help-seeking behavior being the main channel of reporting (70% of all calls). We also show that help-seeking behavior increased the most during the strictest mobility reduction period, and in provinces with the most stringent mobility reduction during confinement. Furthermore, we provide evidence of a significant decrease in detection and protection of victims by both the police forces and the judicial administration over the 9 months period following the pandemic outbreak. Despite the large increase in reported IPV during the pandemic, we estimate that the introduction of lockdown caused a loss of at least 1500 protection orders to victims that would, in the absence of the pandemic, have received it.

Our findings have important policy implications. We provide evidence that the emergency hotline appeared to be a primary channel of help-seeking behavior for women during lockdown but also beyond, while traditional channels of protection (the police) became less accessible/safe for women. As such, we believe that emergency hotlines should be given a central role in the overall detection and protection system of IPV victims and foster a more direct and systematic communication with police forces, women's shelters and/or health workers. A direct consequence of this substitution in reporting is the decrease in judicial protection given to women, which is acquired after filling former complaints to the police forces: significant efforts should be engaged to detect women facing the highest risks of abuse, and to fasten judicial administrative processes related to protection orders' decisions. We also show that while help-seeking behavior is the main channel of reporting, third party reporting represented a key channel in the immediate aftermath

[4] There are 17 regions (Autonomous Communities), 50 provinces and 2 autonomous cities (Ceuta and Melilla)

of the implementation of the lockdown. As such, we argue that awareness campaigns directed to the general population aiming to raise concerns and empathy towards IPV victims might be an efficient mechanism of increasing detection of IPV victims through third parties.

Our paper is organized as follows: Section 2 provides some institutional background on male-based violence policies in Spain, as well as an overview of the chronology of the quarantine measures implemented since the COVID-19 pandemic outbreak. In Section 3 we analyze the effect of the introduction of the lockdown on changes in calls to the emergency hotline, according to the type of caller. In Section 4 we investigate differences in effective mobility reductions across provinces since the introduction of the lockdown. In the last section, we exploit information on MBV related complaints and judicial protection to explore the relationship between these different reporting channels and protection of IPV victims.

2 Background

2.1 Institutional background

The Spanish law recognizes male-based violence as a specific type of abuse with disastrous consequences, and its detection has been given increased attention in the political agenda in the past decades.

In 2004, Organic Law 1/2004 of Integral Measures against Gender Violence marked an important turning point in defining an integrated system of protection and prevention of victims of male-based violence[5]. This system revolves around three main pillars, which intervene at different stages of the detection and/or protection of MBV victims: (1) the Government Delegation for Gender-based Violence (DGVG), which is responsible for coordinating actions against male-based violence between the different institutions and regions. Resources managed by the Delegation include the national emergency hotline for victims of gender-based violence (016) and emergency shelters[6]; (2) the Ministry of the Interior through the State police, which is responsible for the investigation and persecution of cases of gender violence. Police complaints for abuse against women and girls, including MBV, are recorded separately from other crimes and offenses to allow

[5] In 2014, UN Women recognized Spain's Organic Law against Gender Violence as one of the most effective laws worldwide in combating and eliminating sexist violence https://violenciagenero.igualdad.gob.es/pactoEstado/docs/FolletoPEVGengweb.pdf

[6] Another resource managed by the Delegation is the ATENPRO system, a telephone device assistance offered to victims of gender-based violence which enables them to be in permanent contact with a support center. The Delegation also examines applications for specific allowances for victims of abuse

for proper tracking and intervention[7]; (3) The Ministry of Justice, through the specialized Courts for Violence against Women which have jurisdiction for both criminal and civil cases on acts that constitute crimes of gender violence (Ministerio de la Presidencia Relaciones con las Cortes e Igualdad, 2019). In order to have a global view of the effect of the pandemic outbreak on the overall system of detection and protection of MBV victims in Spain, it is important to investigate the extent to which the pandemic affected victims' detection and protection at different stages: first, we focus on detection through the emergency hotline, which became a key resource during the pandemic. Second, we look at MBV related complaints filed to the police, and finally we study the judicial decisions regarding protection of victims (protection orders).

Emergency hotline: The national emergency hotline was created in 2008. This service offers information, judicial advice, and psychological support for any type of violence against women (psychological, physical, sexual). Calls to the number are free, available 24 hours a day, every day of the year and in 53 different languages. Some calls made can be rerouted directly by the counsellors to the emergency number 112 - which handles all types of emergencies: from health to security. However, there is no direct and systematic communication between the hotline and the police, women's shelters, or health centers. Rather, hotline counsellors redirect women to these "traditional" institutions after assessing the severity of the situation. Approximately 70,000 calls are made to the hotline every year.

Complaints: Filing a complaint for MBV is a necessary step for any type of judicial protection of the victim. The great majority of complaints are filed through the Police forces, which then issue a report attesting the facts leading to the complaint[8]. Complaints for domestic abuse are mostly filed by victims themselves (in 70% of all cases) but can also be filed by the police without prior complaint - for instance after an intervention with clear signs of MBV - and in fewer cases by individuals related to the victim or institutions aware of the crime.

Complaints cannot be filed through calls or online - victims need to go to a police station. Each year, approximately 160,000 complaints for abuse against women are reported, 90% of the aggressors being either partners or ex-partners. This represents more than 3 times the number

[7] The state police also manages the VioGèn system, which is one of the Spanish policies to fight violence against women that is considered most advanced in the world. This system aims at monitoring the risk of women that already faced male-based violence (generally domestic), by exploiting real-time information from social services, health services and the police on either the victim or her potential aggressor. Unfortunately, this information is only available at yearly level and not for all regions.

[8] It is also possible, although much less frequent, to file a complaint to the judge directly

of calls to the emergency hotline. In that sense, we consider reporting of abuse to the police[9] as the main channel of detection for victims of gender-based violence in Spain. Reporting to the police can be a costly decision for women for several reasons: lack of trust in police forces (Amuedo-Dorantes and Arenas-Arroyo, 2019), fear of retaliation, shame... Reporting the abuse might have been even more challenging for women during the pandemic, as their movements could be more easily monitored by the aggressor or for fear of the virus. Therefore, we expect that women might have turned to other channels (the hotline) to seek help during the pandemic. ***Protection orders:*** After a complaint is filed, the police might encourage women to file an application for a protection order – this is usually done at the police station right after filling in the complaint (as with complaints, third parties related to the victims, NGO's or public institutions aware of the crime can also request a protection order). Protection orders are court judgements taken in situations where there is both strong evidence of gender-based violence, and which present a clear risk for the victim. A judge can order the protection of a victim through the adoption of civil measures (conditions regarding parental authority, enjoyment of the dwelling…) or criminal measures (prison, prohibition to approach the victim, return to her residence and prohibition to communicate). Each year, approximately 40000 protection orders applications are filed to the Courts of Violence against Women, 70% of them being granted.

2.2 Timeline of quarantine measures during COVID-19

The national lockdown was implemented on March 15th in all regions in Spain under very strict conditions. While most countries maintained authorization to go outside for physical activity or stroll around even at the highest peak of the pandemic (particularly for families with children), in the Spanish case this was not allowed until two months later during the 1st phase of the deconfinement measures.

The quarantine was progressively relaxed from early June, after nearly two months of strict lockdown. Progressive deconfinement measures were implemented at different times and under different conditions across the Spanish territory. The most strongly hit provinces[10] remained in the strictest phase of the quarantine longer than the others. By mid June, all Spanish regions were imposed milder social distancing restriction, mobility between provinces was restored, cultural

[9] While in some rare cases complaints are filed to the Judge directly, for simplicity we will refer to IPV related complaints as police complaints as the great majority of them are filed through the police forces.
[10] There are 52 provinces and 17 autonomous communities in Spain and two autonomous cities (Ceuta and Mellila). Autonomous communities (or CCAA) correspond to the highest level of local administration (regional).

events could take place, bars and restaurants could open… This "new normality" phase lasted over the summer, up to October. On the 20th of that month, curfew was established as a response to the subsequent waves of the pandemic, and was not relaxed until April 2021 (See section **A.a** for further details on the chronology of social distancing measures in Spain between March 2020 and March 2021).

3 The effect of the national-level quarantine measures on victim and third-party reporting

In the context of the quarantine, reporting domestic abuse to the police might have been even more challenging for victims (because of control tactics used by partners) or third parties (for fear of going outside). In this context, reporting of abuse through the hotline might have appeared as an "easier" solution to seek assistance. In this section, we explore the effect of the implementation of the national-level quarantine measures on victims' and third-party reporting through the emergency hotline, both on average and overtime between March 2020 and December 2020.

3.1 Data: Calls to the emergency hotline

We use as our indicator of help-seeking behavior of IPV victims the number of calls to the emergency hotline managed by the Government Delegation for Gender-based Violence (DGVG). Data is available at the province level, for each year and month between 2008 and 2020, according to the caller: the victim, or a third party[11].

Calls reported in the data specifically relate to gender-based violence – calls made for general information purposes are not included. While data is available from 2008 to 2020[12], we restrict our analysis to the period 2013-2020 as some of the early years (from 2008 to 2012) were affected by a number of shocks (communication campaigns on hotline use, economic crisis…) likely to affect the comparability of these years with 2020.

Our final dataset consists of 4,992 cells defined at the province, month and year level between January 2013 and December 2020. We define 3 categories of outcomes depending on the type of

[11] Which can either be family members, neighbors, friends. Unfortunately, the available data does not allow us to precisely identify which third party is making the call.
[12] We do not include in our analyses data from 2021, which is only available for the months of January and February.

caller (all callers, victims, third parties) and construct for each category two outcomes of interest: the number of calls per 100,000 inhabitants and the log of the number of calls.

- $CallsHab^{tot}_{p,y,m}$ is the total number of calls per 100,000 inhabitants and $Log(Calls^{tot}_{p,y,m})$ the log number of calls in province *p*, year *y* and month *m*.
- $CallsHab^{V}_{p,y,m}$ is the number of calls *made by victims themselves* per 100,000 inhabitants and $Log(Calls^{V}_{p,y,m})$ the log number in province *p*, year *y* and month *m*.
- $CallsHab^{Th}{}_{p,y,m}$ is the number of calls *made by third parties themselves* per 100,000 inhabitants and $Log(Calls^{Th}_{p,y,m})$ the log number in province *p*, year *y* and month *m*.

The yearly number of calls made to the emergency hotline in Spain between 2013 and 2020 ranges from 65,000 to 85,000, with the highest number of calls registered in 2015 (**Fig A 1**): this is equivalent to yearly calls per 100,000 inhabitants ranging from 138 to 180. It is important to note that the pandemic outbreak in early 2020 coincides with a sharp increase in both the number of calls made by victims and third parties, thereby breaking the overall decreasing trend in calls to the hotline registered since 2015. Over the years considered, the share of third-party calls remained stable around 30%.

3.2 Identification strategy

We exploit province-monthly level information on the number of calls to the hotline between 2013 and 2020 and formally quantify the effect of the pandemic outbreak on reported MBV using a difference-in-differences framework, where our "treated" year is the post-pandemic outbreak year (2020) and our "control" years are the pre-pandemic years from 2013 to 2019. The pre-policy (lockdown) period that we consider for each control and treatment year are the months of January and February; the post-policy months are those from March to December. **Fig A 2** in the appendix displays changes in calls relative to February for both the treatment and control groups. Our identification strategy relies on the assumption that in the absence of the quarantine shock, seasonality in detection of male-based violence throughout 2020 would have been similar to that of previous years (common trend assumption). We estimate average effects using a difference in differences specification (DiD) and investigate the dynamics of this effect in the short- and medium-term using event study models (EVS), which also allows us to check for pre-trends.

In all specifications presented below, standard errors are clustered at the province level to account for the autocorrelation of residuals within each province overtime.

Estimation of the average effect of the implementation of quarantine between March and December 2020

Our econometric specification for the estimation of the average effect of the lockdown introduction on MBV reporting is the following:

$$\textbf{\textit{Eq 1}}.\ CallsHab^{j}_{p,y,m} = \alpha + \beta_0 D_{y=2020} + \beta_1 D_{m\geq 3} + \beta_{did} D_{y=2020} D_{m\geq 3} + \gamma_p + \gamma_y + \gamma_m + \epsilon_{p,y,m}$$

With *j= {Tot; V; Th}* the indicator for total, victim, or third-party calls per 100,000 inhabitants. $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March[13], and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of the lockdown introduction on the 15th of March 2020 (and onwards) with respect to the previous years. We include province-level fixed effects to account for time-invariant characteristics that are likely to affect IPV reporting through the hotline (demographic characteristics, population density, etc.); year fixed-effects to account for common time trends across provinces (*e.g* higher level of calls during economic recessions); month fixed effects to control for seasonality changes in IPV reporting within a given year[14].

We run the same regression on the log of calls controlling for population size ($Pop_{p,y}$) which allows us to interpret regression coefficients in terms of percentage change in calls.

$$\textbf{\textit{Eq 2}}.\ Log\left(Calls^{j}_{p,y,m}\right) = \alpha + \beta_0 D_{y=2020} + \beta_1 D_{m\geq 3} + \beta_{did} D_{y=2020} D_{m\geq 3} + \gamma_p + \gamma_y + \gamma_m + Pop_{p,y} + \epsilon_{p,y,m}$$

In alternative specifications, we add additional time-varying controls likely to affect IPV in each province: total unemployment rate, male and female unemployment rate, population density.

Estimation of the dynamic effect of quarantine implementation between March and December 2020

After estimating the average effect of the implementation of the quarantine between March and December 2020, we explore the timing of the effect on reported DV in the short- and medium-term by running event study models on the log number of calls to the hotline on all seven years of reporting data (2013-2020):

[13] We consider an intensity of treatment of 0.5 for the month of March since lockdown was implemented on the 15th.

[14] $D_{m\geq 3}$ partially controls for seasonality by isolating, respectively, January-March from April-December. The inclusion of month fixed effects allows to control for seasonality within each of the subgroups defined by $D_{m\geq 3}$.

$$\textit{Eq 3.}\ Log(Calls^{j}_{p,y,m}) = \alpha + \beta_0 D_{y=2020} + \sum_{i=-2}^{9} D_{m=3+i} + \sum_{i=-2}^{9} \eta_i D_{m=3+i} D_{y=2020} + \gamma_p + \gamma_y + Pop_{p,y} + \epsilon_{p,y,m}$$

We specify dummies ($D_{m=3+i}$) for the two months of lag before lockdown implementation, and 9 months of leads up to December to capture the long-term effect of the quarantine on DV reporting. For each of these dummies, η_i captures the additional effect of the pandemic outbreak on the considered month in 2020, relative to control years (2013-2019).

3.3 Results

Regression results for our three variables of interest (total calls, calls by victims, calls by third parties) expressed as per 100,000 inhabitants and log-transformed are presented below:

Table 1. Impact of lockdown introduction on monthly MBV reporting through emergency hotline – difference-in-differences estimates.

	(1) Log of calls	(2) Log of calls by victims	(3) Log of calls by third parties	(4) Total calls per 100,000	(5) Calls by victims by 100,000	(6) Calls by third parties by 100,000
$\boldsymbol{D_{y=2020} D_{m\geq 3}}$	**0.390*** **	**0.380*** **	**0.414*** **	**4.446*** **	**2.983*** **	**1.463*** **
	(0.026)	(0.030)	(0.035)	(0.347)	(0.270)	(0.116)
$\boldsymbol{D_{m\geq 3}}$	0.040+	0.034	0.027	0.358+	0.201	0.157+
	(0.023)	(0.023)	(0.029)	(0.187)	(0.133)	(0.086)
$\boldsymbol{D_{y=2020}}$	0.039	0.065*	-0.025	0.447+	0.490*	-0.043
	(0.025)	(0.030)	(0.030)	(0.236)	(0.206)	(0.070)
Constant	3.838***	3.438***	2.712***	10.470***	8.260***	2.210*
	(0.024)	(0.028)	(0.028)	(2.879)	(1.918)	(1.024)
Observations	4990	4984	4949	4992	4992	4992
R^2	0.214	0.176	0.131	0.249	0.209	0.159
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March[15], and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years. Standard errors are clustered at the province level.
Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

We estimate that the introduction of the first quarantine policies following the pandemic outbreak is responsible for an average increase in IPV reporting of 47.7%[16] between March 2020 and December 2020 compared to previous years over the same period. This effect corresponds to an

[15] We consider an intensity of treatment of 0.5 for the month of March since lockdown was implemented on the 15th.
[16] Exp(0.39)-1

additional +26345[17] calls compared to the control years (see **Table A 2** in appendix for reference values), and is equivalent to an additional +4.5 calls per 100,000 inhabitants each month (Specification 4).

Estimation results from specifications 4 to 6 allow us to disentangle precisely the extent to which help-seeking behavior of IPV victims on the one hand, and reporting by third parties on the other hand, contributed to the increase in reported DV that we document. We find that 67% of the increase in reported DV between March and December 2020 can be attributed to reporting by victims' themselves, against 33% for third party calls[18]. Furthermore, the composition of reported DV in terms of who is reporting the violence changes only very slightly after lockdown (38% increase for victims' reporting against 41% for third parties). Our results are opposite to those in Ivandic, Kirchmaier and Linton's, which report that the increase in reported DV hotline calls in the Greater London area during quarantine steamed exclusively from third parties reporting. While the latter provide indirect evidence that this does not reflect a pure reporting effect[19], we are able to directly rule out this hypothesis in the context of an entire country (Spain), over a 9-month period.

Results are robust to alternative specifications in which we control for population density as well as the local unemployment rate (See **Table A 3, Table A 4** and **Table A 5** in the appendix)[20].

We next proceed by investigating the dynamics of the effect of lockdown overtime and according to the type of caller. Event study plots of the effect of quarantine implementation in March 2020 provide suggestive evidence of the existence of common pre-trends in the first two months in the affected and unaffected years for all calls (**Fig A 3**), and for calls by victims and third parties (**Figure 1**). The first red dotted line indicates the month of implementation of the national quarantine measures in Spain (15th of March), which should be considered as a partially treated month. The second grey dotted line marks the beginning of the more "relaxed" period in terms of mobility restrictions, which lasted over the summer months up until mid-October. The last dotted

[17] Which is found by multiplying the estimated % effect (47.7%) by the mean of the number of calls in the previous years 55230, see Table A 2

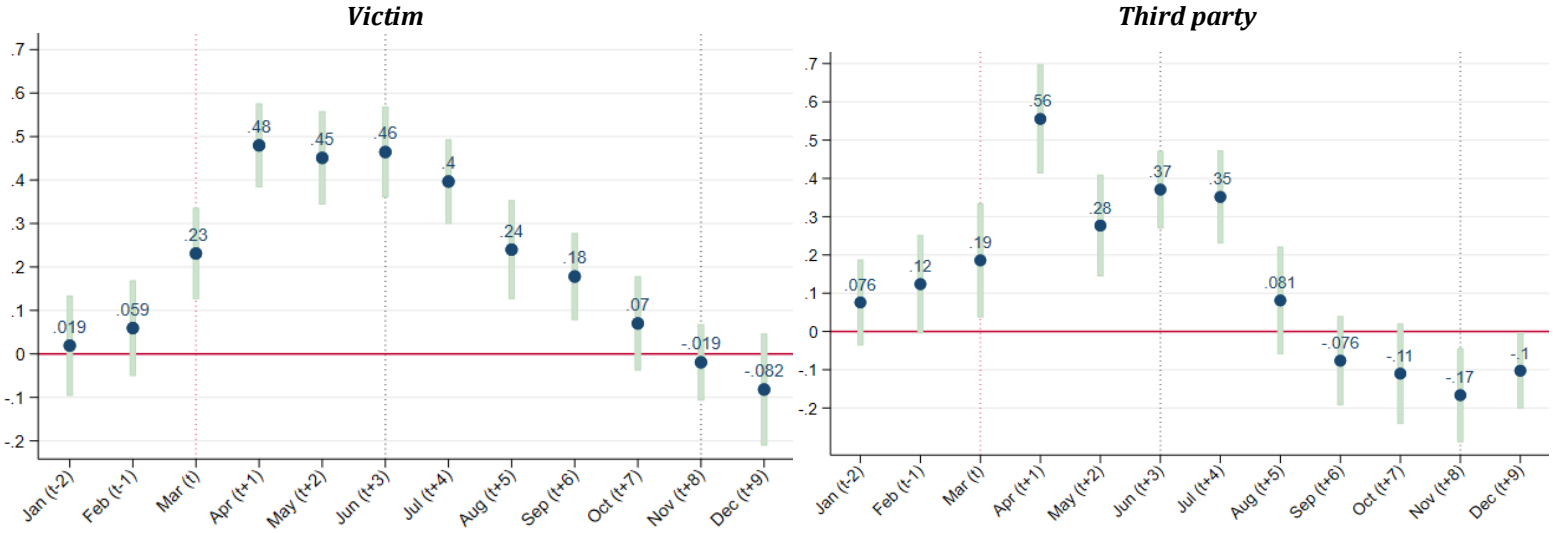
[18] To obtain the share of additional DV reporting due to help-seeking behavior, we divide the estimated additional calls by victims (specification 5) by the additional total calls (specification 4).

[19] Using information on third party reporting during working hours and evening hours

[20] We note that coefficient estimates regarding the effect of unemployment rate on reported IPV should be interpreted with caution given the endogeneity of this variable in our model. The stability of our results to the introduction of this variable shows, however, that changes in reported IPV are driven by other channels than the sole economic instability generated after the pandemic.

grey line marks the beginning of a second wave of restrictions, which included a national curfew at 10.00pm and the closure of bars and restaurants.

Figure 1. Impact of lockdown introduction on MBV reporting through emergency hotline, according to the type of caller. Event study estimates: log models



For both third parties and victims, we find that the highest increase in reported abuse occurred during the strict mobility reduction period – between March and June. This is suggestive of the importance of strict mobility reduction in the documented increase in reported MBV (we further investigate the role of mobility reduction on IPV prevalence in the next section). We also observe that the effect of the introduction of the quarantine persists beyond the duration of the confinement for both third parties and victims (which is indicated in the figures by the red dotted line and the grey dotted line). This persistent effect could be explained by the persistence of worse economic conditions, an inertia of violence, whereby violent conflict in the household is more likely to occur once previous violent events have already taken place, even when the conditions that gave rise to the previous violent aggressions are no longer the same. This could also reflect a "delay" in reporting, whereby victims who were not able to contact the hotline while being permanently with the violent partner are finally able to report the domestic abuse after the confinement was lifted.

We find, on the other hand, that the *dynamics* of the effect of the introduction of lockdown is different according to the type of caller: the strongest effect of the lockdown on third party reporting was registered in April 2020, which is the first fully treated month (75% increase[21]). This effect is reduced, in terms of its magnitude, in the following two months of strict quarantine (May and Early June)and becomes non statistically different from 0 from August onwards. On the

[21] Exp(0.56)-1

other hand, the increase in victims' reporting remained at 58%[22] in April, May, and June, and only started to (slowly) lose its magnitude in the "relaxed summer" phase to become non-statistically different from 0 in November 2020 (against August 2020 for third party reporting). The spike in third party reporting immediately after lockdown could steam from an awareness effect, whereby neighbors - being always at home - could now witness/hear domestic violence episodes that they would otherwise not have noticed. This could also reflect a strong increase in MBV itself immediately after lockdown. While we are not able to precisely identify the drivers of these differences, our results are nonetheless insightful as they provide evidence that third party reporting might be very sensitive to large shocks such as the quarantine implementation, which makes it a key player in the overall system of detection and protection of IPV victims.

4 Heterogeneity analyses across provinces: the role of *effective* mobility reductions and pre-pandemic characteristics

To shed light on the possible mechanisms underlying the reported DV's response to the implementation of the quarantine measures, we ran heterogeneity analyses for subgroups of provinces likely to experience significant differences in levels of IPV reporting during the pandemic. We explore three categories of factors at the province level: (1) effective mobility reductions during confinement; (2) pre-pandemic level of IPV incidence, which we measure using quarterly police data, protection orders, and calls to the emergency hotline (3) pre-pandemic demographic and labor characteristics, measured by population and share of essential workers.

4.1 Data

Effective mobility reductions during confinement

One key hypothesis of this paper and in the overall literature on the effect of the pandemic on the incidence of IPV is that mobility restrictions, and the particularly unprecedented, forced coexistence induced by quarantine policies increased the risks faced by women and children to suffer violence from men at home. In order to quantify the extent of this relationship, we use information from the COVID-19 mobility database of the National Institute of Statistics (INE) on the share of the mobile population at the province level, between March 2020 and June 2020 and during a reference week in November 2019. This allows us to measure the stringency of the mobility reduction based on real mobility data at the local level, which overcomes possible data

[22] Exp(0.46)-1

limitations of existing stringency measures based on Google Mobility Data or the Oxford Coronavirus Government Response Tracker (OxCGRT)[23].

We use this information to construct a continuous indicator of stringency of the confinement at the province level for each province ($String_p$), which is equal to the average decrease in mobility between March and June 2020 compared to the reference period. We also construct a binary indicator which classifies provinces into high and low mobility reduction (or stringency) according to the median value of $String_p$. More details on the mobility data and the indicators used are presented in appendix **section A.b.**

Fig A 4 in the appendix section displays the stringency indicators $String_r$ defined at the regional level (CC.AA), compared to the pre-pandemic (November 2019) levels. While formal mobility was equally restricted on the entire Spanish territory during the initial phase of the lockdown measures, this map shows that the *effective* mobility reduction differed across Spanish autonomous communities. Descriptive statistics on the average mobility reduction in Spain (**Table A 1**) and the evolution of the number of calls in high and low mobility reduction provinces (**Fig A 5**) are provided in the appendix**.**

Differences in effective mobility reductions during confinement could steam from pre-pandemic province characteristics (such as demographics, labor market structure, etc.), from differences in the evolution of the pandemic across the Spanish territory (with mobility being more restricted in the most hit provinces, either because of a stronger enforcement of the confinement measures by the local police or because of a stricter quarantine compliance by the population due to the fear of being infected/spreading the virus) or from differences in labor market characteristics (share of essential workers, unemployment rate). Using an OLS regression of the province monthly decrease in mobility (**section A.b**) on population density, COVID-19 incidence, and the local unemployment rate during confinement, we find that these variables do not significantly explain differences in mobility reductions across provinces[24]. We therefore consider differences in mobility reductions during confinement as a separate potential factor of IPV reporting beyond population density and labor market characteristics.

[23] Oxford Coronavirus Government Response Tracker (OxCGRT)

[24] The effect of the share of essential workers, which is time invariant for each province, is captured by provinces' fixed effects.

Pre-pandemic levels of IPV incidence

We now consider whether social distancing increased violence among provinces with the highest IPV incidence before the pandemic outbreak (intensive margin) or caused an increase in violence in provinces with no or a low history of domestic abuse (extensive margin). Differences in pre-pandemic IPV incidence are measured using IPV-related complaints at the province-quarter of year level filed by either the victim or the police. We also consider protection orders at the province and quarter of year level. Plots of the evolution of IPV complaints and protection orders per 100,000 inhabitants are presented in the appendix (**Fig A 6** and **Fig A 7).** For each measure of IPV incidence per 100,000 inhabitants, the threshold used to differentiate provinces into the high or low subgroup is the median of the province average between 2015 and 2018 **(Fig A 8**)

Pre-pandemic demographic and labor market structure

Only workers from essential sectors were allowed to commute during the strict quarantine period. We could therefore expect the share of essential workers to affect effective mobility and employment during quarantine, which in turn is likely to affect IPV incidence (less time spent at home, less economic uncertainty) and reporting (neighbors less likely to witness/hear the abuse). Population density, on the other hand, is likely to affect third party reporting through the probability of witnessing/hearing the abuse.

We investigate the role of these pre-pandemic labor market and demographic characteristics on changes in reported IPV by differentiating provinces with respect to the share of essential workers and population density. Our measure of provinces' share of essential workers is taken from the INE, as defined by the Royal Decree 10/2020 based on employment composition of provinces on the 29th of March 2020. We take as reference value for the pre-pandemic density of population the density as of the 1st of January 2019 (INE). We define as high density provinces those whose density in 2019 belongs to the top 25th quantile of the distribution of population density so as to identify highly populated areas. Provinces are categorized into the high or low share of essential workers categories based on the median value of the share of essential workers.

4.2 Identification strategy

We estimate the additional effect of the quarantine on reported IPV based on provinces' characteristics using data from 2013 to 2020, with a model of the form:

$$\textbf{\textit{Eq 4.}}\ Log(Calls^{i}_{g,m,p,y}) = \alpha_0 + \beta_1 D_{m\geq 3} + \beta_2 D_{y=2020} + \beta_{did} D_{m\geq 3} D_{y=2020} + \delta_{ddd} D_g D_{m\geq 3} D_{y=2020} + \gamma_p + \gamma_m + \gamma_y + \epsilon_{w,p,y}$$

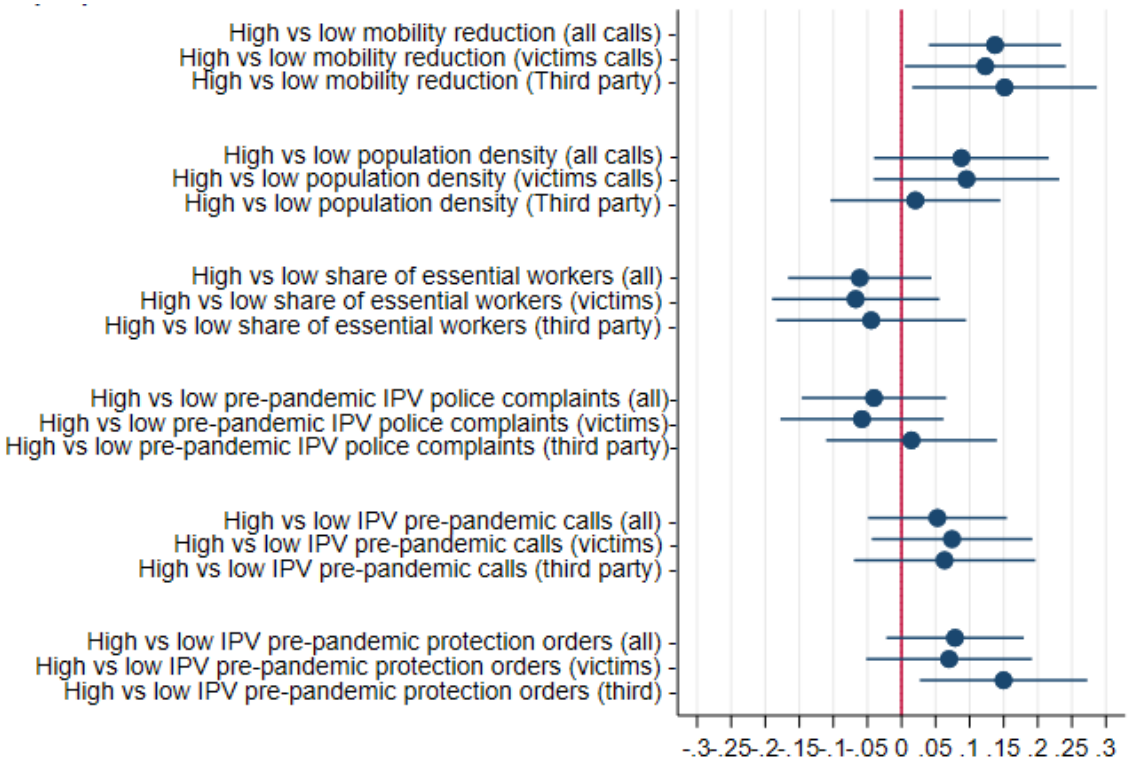
g is the criteria that we use to distinguish our provinces (pre-pandemic IPV incidence, density, etc.). As before, $Log(Calls^{i}_{g,m,p,y})$ is the log number of reported IPV episodes for the type of caller i (all callers, victims or third parties) when considering differences between provinces related to g. D_g is a dummy indicator dividing the provinces into the high and low subgroups: the group of interest D_g=1 (*e.g.* provinces with high mobility reduction) and the comparison group defined by D_g=0 (*e.g.* provinces with low mobility reduction). β_{did} measures the average effect of the lockdown on changes in reported DV. δ_{ddd} measures the *additional* effect of lockdown for provinces of type D_g=1, compared to the rest of provinces. The total effect of the introduction of lockdown on the subgroup D_g=1 is equal to β_{did}+δ_{ddd}. To account for the fact that mobility reduction during confinement is likely to be partially affected by population density, we control for this variable in the heterogeneity analyses according to the degree of mobility reductions[25].

4.3 Results

Results of the heterogeneity analyses are presented in the figure below, which plots the value of the δ_{ddd} coefficients (**Eq 4**)

[25] The share of essential workers, which also likely affected differences in mobility across provinces, is accounted for by province fixed effects in the regression.

Figure 2. Additional effect of the quarantine measures according to mobility reduction and pre-pandemic characteristics, in % change (log models)



We first analyze whether pre-pandemic IPV incidence affects differences in post-quarantine IPV reporting between provinces. While coefficients are not statistically different from 0 for victims' reporting and overall reporting, we see that provinces with higher pre-pandemic protection orders per 100,000 inhabitants exhibit a higher increase in reported IPV after the lockdown. More specifically, the effect for third party reporting is 15% higher. As protection orders represent one of the most advanced forms of victims' protection and are usually dictated for extreme situations of repeated domestic violence, we interpret this result as increases in IPV reporting being stronger in provinces with a historically higher incidence of "extreme" abuse (intensive margin). We find, on the other hand, that the effect of the quarantine on calls is not statistically different between provinces with different levels of pre-pandemic IPV related complaints. Overall, we do not find evidence that the increase in IPV reporting was driven by provinces with the lowest IPV incidence before the pandemic - contrary to previous findings in the literature (Leslie and Wilson, 2020).

We find that provinces with the most stringent mobility reduction during the confinement experienced the strongest increase in reported IPV through calls (+16%) compared to low mobility reduction provinces. This supports our hypothesis that mobility reduction is a key driver of the increase in DV documented in this paper. To check that mobility reduction estimates do not capture the same effect as pre-pandemic IPV incidence - i.e., that provinces with higher mobility

reduction are not the same as those with higher IPV incidence -, we measure the correlation between average mobility reduction during confinement and pre-pandemic protection orders per 100,000. We find a positive but weak correlation[26] between mobility and pre-pandemic protection orders, which suggests that pre-pandemic levels of protection orders and mobility are distinct factors leading to an increase in IPV reporting.

Overall, we find that effective mobility reductions and previous history of IPV incidence are the main drivers of differences in IPV reporting across provinces. This suggests, in turn, that violence is highly sensitive to the time spent at home and that violence might have increased at the intensive margin in the aftermath of the outbreak of the pandemic.

5 Substitution between channels of reporting and victims' protection

In this section, we aim to understand whether victims are adapting their help-seeking behavior to the constraints imposed by the pandemic and, if so, we want to measure the consequences of this substitution on the overall protection of victims. More specifically, in this section we: (1) discuss the extent to which the sharp increase in IPV reporting through hotline calls by victims steams from a substitution effect between hotline reporting and reporting to the police (2) estimate the loss in victims' protection caused by the pandemic outbreak, as measured by protection orders.

5.1 Data

To do so, we exploit information from the *Consejo General del Poder Judicial (CGPJ)* on gender-based violence complaints and protection orders granted by the Courts of Violence against Women. Data is available at the province level between 2009 and 2020 for each quarter of year. In this part of the analyses, we do not exclude the 2009-2013 years in order to improve the estimation of the seasonality of IPV through quarters of year fixed effects.

5.1.1 Complaints

We exploit quarterly-province level data from the CGPJ on the number of complaints for gender-based violence between 2009 and 2020. This information is available for complaints filed by victims, or by third parties (mainly the police). Between 120000 and 180000 complaints for gender-based violence were filed each year between 2009 and 2020, 70% of them being filed by victims themselves (**Fig A 6**). For each province *x* quarter *x* year cell we construct two main

[26] Pearson correlation coefficient of 0.1567

indicators, which we measure both in log and per 100,000 inhabitants: the number of complaints filed by victims, and the total number of complaints.

Although filling a complaint is the key step for any protection from the aggressor by the police or the judicial system, only a minority of victim discloses present or past abusive relationship to the relevant authorities. According to the 2019 Violence Against Women Spanish survey, only 22% of victims of abuse from partners or ex partners reported the violence to the police or the Courts of Violence against Women (Igualdad, 2019). This large under-reporting of violence against women can be driven by several channels - lack of trust in police forces (Amuedo-Dorantes and Arenas-Arroyo, 2019), fear of retaliation, shame... (Igualdad, 2019) - and likely increased with the quarantine policies. Because victims' movements could be monitored much easier by aggressors, or for fear of going outside during the pandemic.

5.1.2 Protection orders

We use as measure of the protection of MBV victims by the Spanish state the number of applications for protection orders, and the protection orders granted by the Courts of Violence against Women. Looking at protection orders' applications together with protection orders granted allows us to compare help-seeking behavior (application for protection) with effective protection. Information is also taken from the CGPJ and is available at the province and quarterly level between 2009 and 2020. For each province *x* quarter *x* year cell we construct two main indicators, which we measure both in logs and per 100,000 inhabitants: the number of protection orders *filed in* (applications), and the number of protection orders *granted* by the judge.

Each year, approximately 40000 protection orders applications are filed to the Courts of Violence against Women, 70% of them being accepted (**Fig A 7**). Delays in court's decision are short: Once the application has been received, an emergency audience is to be planned within 72hours. As such, protection orders granted during a given quarter were filed in for application during the same quarter (no delay)[27].

5.2 Methodology

To formally investigate the effect of the pandemic on the protection of MBV victims, we also use a difference-in-differences framework with 2020 our treated year and 2009-2019 our control

[27] While some judicial delays had been extended/temporarily removed during the state of the emergency, this was not the case for protection orders. *Real Decreto Ley 16/2020, de 28 de abril*

years. Our post period is defined based on the quarter of years. More specifically, we consider the first quarter of year (from January to March) as the pre-treatment period. The subsequent quarters of years from April to December make our post-treatment periods.

Effect of the introduction of lockdown on complaints and protection orders

We estimate the effect of the quarantine implementation on complaints and protection orders using the following difference-in-difference equations:

$$\textbf{Eq 5.}\ Y^{j}_{p,y,q} = \alpha + \beta_0 D_{y=2020} + D_{q\geq 2} + \beta_{did} D_{y=2020} D_{q\geq 2} + \gamma_p + \gamma_y + \gamma_q + Pop_{y,p} + \epsilon_{p,y,q}$$

$$\textbf{Eq 6.}\ Log(Y^{j}_{p,y,q}) = \alpha + \beta_0 D_{y=2020} + D_{q\geq 2} + \beta_{did} D_{y=2020} D_{q\geq 2} + \gamma_p + \gamma_y + \gamma_q + \epsilon_{p,y,q}$$

$Y^{j}_{p,y,q}$ is the number of complaints related to gender-based violence (total and filed by the victim) or the number of protection orders (applied for and granted) in province *p,* quarter of year *q,* and year *y.* We explore the effect of the introduction of the lockdown on the outcomes expressed as per 100,000 inhabitants (Eq 5) and log transformed (Eq 6). In both models, $D_{q\geq 2}$ is a dummy variable equal to 1 from the 2nd quarter to the last quarter of each year. Our coefficient of interest is the β_{did} which captures the effect of the introduction of lockdown on IPV related complaints and protection orders. We include province fixed effects to account for time invariant characteristics regarding the local system of protection of victims or the prevalence of violence at the local level, quarters of years fixed effects to capture the seasonality of IPV throughout the years, and years fixed effect to account for time trends affecting both IPV prevalence (economic context) and protection (laws). Standard errors are clustered at the province level.

Substitution between complaints to the police and reporting to the hotline

To gain insights on the possible substitution between reporting to the police and reporting to the hotline, we also run an event study model at the quarterly level for both the total calls and total complaints per 100,000 inhabitants, with the first quarter of year as baseline period and the subsequent quarters of years as lead periods.

$$\textbf{Eq 7.}\ ReportHab^{j}_{p,y,q} = \alpha + \beta_0 D_{y=2020} + \sum_{i=0}^{3} D_{q=i+1} + \sum_{i=0}^{3} \eta_i D_{q=i+1} D_{y=2020} + \gamma_p + \gamma_y + \epsilon_{p,y,m}$$

$ReportHab^{j}_{p,y,q}$ refers to the number of reported abuses through the hotline (resp. to the police) per 100,000 inhabitants. $D_{q=i+1}$ are dummies capturing quarter of years fixed effects. η_i are our coefficient of interest: they capture the change in reporting (to the hotline or the police) in 2020 for a given quarter or year, compared to previous years. Standard errors are clustered at the province level.

5.3 Results

Regression results on the effect of the lockdown introduction on complaints and protection orders are presented in the appendix.

Complaints: We find a positive but non-significant effect of lockdown introduction on complaints filed for violence against women (specifications 1-2 and 5-6, **Table A 7**). Event study estimates (Eq 7) of the effect of the lockdown on changes in reporting to the hotline and to the police per 100,000 inhabitants (**Fig A 10**) indicate a positive increase through both channels of reporting at Q2, between 3 to 6 months after lockdown implementation, when strict mobility restrictions were lifted. At Q3, coefficient estimates compensate each other (-5.2 and 5.2), which might suggest that victims did not durably change their reporting practices and reverted back to reporting to the police in the long term. However, interpretation of these results should be taken with care given the non-significance of EVS estimates relative to complaints per 100,000 inhabitants.

Protection orders: When looking at the effect of the pandemic outbreak on judicial protection (Table A 7, Specification 8), we find a significant decrease in protection orders granted by the courts after the implementation of the lockdown in 2020 compared to control years. More specifically, we find that the quarantine measures introduced at the very end of the 1st quarter of 2020 caused a decrease of -3.38 protection orders application per 100,000 inhabitants[28]. This implies a "loss" of at least 1586 protection orders to victims that would, in the absence of the pandemic, have received it[29]. We find that the decrease in protection orders granted by the Court (Specification 4) is of similar magnitude as the decrease in *applications* for protection orders (Specification 3), of approximately -10%. As applications are mainly done at the police station at the same time as the complaint, this suggests that the loss of judicial protection steams from a decrease in complaints for cases of abuse filed to the police that would be eligible for protection orders (that is, a decrease in complaints for extreme forms of abuse).

Stability of reporting to the police through complaints on the one hand, and evidence of a decrease in reporting and protection for severe cases of abuse on the other hand, could be explained by a stability of the prevalence of violence against women, a change in the type of violence exerted (decrease in extreme forms of violence), or a decrease in reporting. Given the large increase in abuse reported to the hotline by both victims and third parties that we document in this paper

[28] Quarterly effect of lockdown introduction on *granted* protection orders, multiplied by three (-1.127x3)
[29] This back of the envelope calculation was done by multiplying the yearly effect (-3.38) by the total Spanish population in 2020 (46 937 056), divided by 100 000. Similarly, we find a loss of 2142 applications for protection orders.

and previous evidence on the effect of the quarantine on reported and unreported domestic violence (including of severe forms) in Spain (Arenas-Arroyo, Fernandez-Kranz and Nollenberger, 2020), we argue that our findings unlikely steam from a stability of violence or a relative decrease in extreme forms of abuse. Rather, we argue that the pandemic outbreak and subsequent lockdown measures caused a loss of detection of violence by the police, and that this loss in detection might have been higher for cases eligible for protection orders, which are usually the most extremes forms of abuse: for instance, because the victims of severe abuse might have been even more limited in their movements by the aggressive partner movement were being intensively monitored by the violent partner. Part of the reduction in protection orders granted by judges could also come from slower judicial procedures in the context of the pandemic, during which they mostly had to work from home (as opposed to police forces). This latter channel is however less probable, as protection orders were among the few judicial processes for which legal delay to process the application (72 hours) was not affected by the state of emergency.

Overall, our results point to the need to detect the most vulnerable women whose risks are currently not monitored and are not protected by judicial orders. This is a crucial step in order to engage the necessary means to protect them from extremes forms of violence such as the ones recently witnessed in Spain in the aftermath of the quarantine[30].

[30] https://elpais.com/sociedad/2021-05-21/cinco-asesinatos-machistas-en-cuatro-dias-un-efecto-del-fin-de-las-restricciones.html

6 Conclusion

We estimate that the introduction of the quarantine policies as a response to the pandemic outbreak is responsible for an average increase in IPV reporting of 48% between March 2020 and December 2020, which corresponds to an additional +26,000 calls compared to previous years. Contrary to previous findings, we find that 67% of these additional calls were made by victims' themselves against 33% that were made by third party calls. Additionally, we show that the composition of reported DV in terms of who is reporting the violence only changes slightly after the lockdown. We find that effective mobility reductions and previous history of IPV incidence are the main drivers of differences in IPV reporting across provinces. This suggests, in turn, that violence is highly sensitive to time spent at home, and that violence might have increased at the intensive margin in the aftermath of the pandemic.

Differences with previous estimates could steam from the granularity of the data used (weekly vs monthly), the time frame used (weeks or months during the pandemic and previously), the geographical level and countries analyzed (most of the literature has focused on large and populated cities, usually in the US). They could also steam from differences in the national or local system of detection of IPV victims, which differs significantly across countries and cities.

When looking at the effect of the pandemic outbreak on traditional channels of detection and protection of IPV victims in Spain, we find evidence that reporting through the hotline increased substantially while reporting to the police was greatly reduced compared to the increase in violence that we document. As a consequence, our estimates show that the quarantine caused a "loss" of at least 1500 protection orders to victims that would, in the absence of the pandemic, have received it. Our results point to the crucial need to detect the most vulnerable women whose risks are currently not being monitored and who are not protected by judicial orders, in order to engage the necessary means to protect them from extremes forms of violence such as the ones recently witnessed in Spain in the aftermath of the quarantine.

Bibliography

Amuedo-Dorantes, C. and Arenas-Arroyo, E. (2019) 'Police Trust and Domestic Violence: Evidence from Immigration Policies', *IZA Discussion Paper*, (12721), p. No. 12721.

Anderberg, D. *et al.* (2016) 'Unemployment and Domestic Violence: Theory and Evidence', *Economic Journal*, 126(597), pp. 1947–1979. doi: 10.1111/ecoj.12246.

Arenas-Arroyo, E., Fernandez-Kranz, D. and Nollenberger, N. (2020) 'Can't Leave You Now ! Intimate Partner Violence under Forced Coexistence and Economic Uncertainty', *IZA Discussion Paper*, (13570).

Beland, L.-P. *et al.* (2020) 'COVID-19, Family Stress and Domestic Violence: Remote Work, Isolation and Bargaining Power', *IZA Discussion Papers*, (13332). Available at: https://www.iza.org/publications/dp/13332/covid-19-family-stress-and-domestic-violence-remote-work-isolation-and-bargaining-power.

Berniell, I. and Facchini, G. (2020) 'COVID-19 Lockdown and Domestic Violence: Evidence from Internet- Search Behavior in 11 Countries COVID-19 Lockdown and Domestic Violence: Evidence from Internet-Search Behavior in 11 Countries'. Available at: http://sedici.unlp.edu.ar/bitstream/handle/10915/112276/Documento_completo.pdf?sequence=1.

Campedelli, G. M., Aziani, A. and Favarin, S. (2020) 'Exploring the Effect of 2019-nCoV Containment Policies on Crime: The Case of Los Angeles', pp. 1–49. doi: 10.31219/osf.io/gcpq8.

Card, D. and Dahl, G. B. (2011) 'Family violence and football: The effect of unexpected emotional cues on violent behavior', *Quarterly Journal of Economics*, 126(1), pp. 103–143. doi: 10.1093/qje/qjr001.

Igualdad, M. de (2019) 'Macroencuesta de violencia contra la mujer. Delegación del Gobierno contra la Violencia de Género.', *Ministerio de Igualdad*, (268), p. 341. Available at: https://violenciagenero.igualdad.gob.es/violenciaEnCifras/macroencuesta2015/Macroencuesta2019/home.htm.

Ivandic, R., Kirchmaier, T. and Linton, B. (2021) 'The Role of Exposure in Domestic Abuse Victimization: Evidence from the COVID-19 Lockdown'.

Kumar, A. and Nayar, K. R. (2020) 'COVID 19 and its mental health consequences', *Journal of mental health (Abingdon, England)*, 0(0), pp. 1–2. doi: 10.1080/09638237.2020.1757052.

Leslie, E. and Wilson, R. (2020) 'Sheltering in place and domestic violence: Evidence from calls for

service during COVID-19', *Journal of Public Economics*, 189, p. 104241. doi: 10.1016/j.jpubeco.2020.104241.

Miller, A., Segal, C. and Spencer, M. (2020) 'Effects of the COVID-19 Pandemic on Domestic Violence in Los Angeles', *NBER Working Paper*. Available at: http://www.nber.org/papers/w28068.pdf.

Ministerio de la Presidencia Relaciones con las Cortes e Igualdad (2019) 'Guide to the System of Action and Coordination in Cases of Gender Violence in Spain.', (July), pp. 1–35. Available at: http://www.violenciagenero.igualdad.mpr.gob.es/en/informacionUtil/guia/home.htm.

Perez-Vincent, S. M. *et al.* (2020) 'COVID-19 Lockdowns and Domestic Violence: Evidence from Two Studies in Argentina', *inter- American Development Bank*, pp. 1–48.

Peterman, A. *et al.* (2020) 'Pandemics and Violence Against Women and Children', *Center for Global Development*, (528), p. 43.

Piquero, A. R. *et al.* (2020) 'Staying Home, Staying Safe? A Short-Term Analysis of COVID-19 on Dallas Domestic Violence', *American Journal of Criminal Justice*, pp. 601–635. doi: 10.1007/s12103-020-09531-7.

Ravindran, S. and Shah, M. (2020) 'Unintended consequences of lockdowns: COVID-19 and the shadow pandemic', *NBER Working Paper*.

A. Appendix

a. Chronology of COVID-19 related measures in Spain

State of emergency and strict quarantine (13th of March to 28th of April): the state of emergency was declared on the 13th of March 2020. Two days later, national lockdown was implemented on the 15th of March 2020 under very strict conditions: mobility was limited to essential needs and restrictions included the closure of bars, restaurants, night clubs, movie theatres and theatres, etc. While most countries maintained authorization to go outside for physical activity or stroll around even at the highest peak of the pandemic (particularly for families with children), in the Spanish case this was not allowed until two months later during the 1st phase of the deconfinement measures.

De-escalation phase (28th April to 20th of June) After nearly two months of lockdown under those strict conditions on April 28th, the government announced the Spanish deconfinement plan organized in four different progressive phases. In the 1st and 2nd phase residents could gather in groups of no more than 10 people (15 in the 2nd), practice physical activity or walk outside at specific hours, attend outside events, go to the library, attend places of cult. Museums, theatres, restaurants, shops were also progressively allowed to open under limited capacity (50%). In the 3rd and last phase, restrictions on the duration and time to go outside were lifted. Some restrictions on group gatherings and capacities were maintained to allow for social distancing).

Progressive deconfinement measures were implemented at different times and under different conditions across the Spanish territory. Between the 11th of May and the 18th of May, 70% of all Spanish provinces entered the 1st deconfinement phase, which allowed limited outside activities and small group gatherings. On the other hand, the most strongly hit autonomous communities (Barcelona, Madrid, and part of the region of Castilla La Mancha) remained in the initial phase of the quarantine longer than the others, until the 25th of May (later for some provinces). Additional protective measures were also taken during the progressive deconfinement phases in Barcelona and Madrid, where mobility to and from the two areas was monitored while bars, restaurants, museums, etc. were allowed to open.

"New normality" (20th June to 25th of October): The state of emergency declared on the 13th of March expired the 21th of June, and Spain entered a "new normality" phase, during which all internal and international borders were opened. Maintained restrictions were limited to regulation of maximum occupancy in shops and masks in public when social distancing was not possible. During this new normality phase, some provinces experience significant increases in the

number of cases and deaths due to COVID-19, which paved the way for local implementation of lockdown in Madrid on the 1st of October.

Re-imposition of State of Emergency (25th October to 9th of May): on the 21st of October Spain passed 1 million COVID-19 cases. 4 days later, on the 25th, the government reimposed a state of emergency across all provinces and introduced a national curfew to counter the resurgence. Local authorities were also given powers to ban travel across different regions and to regulate opening of bars and restaurants. On the 9th of May, the state of emergency came to an end: curfew was lifted in all Spanish provinces, together with restrictions on bars and restaurants. Restrictions regarding occupation capacity were relaxed but maintained.

b. Mobility reduction during lockdown

Description of the mobility database

The Spanish National Institute of Statistics (INE) created a mobility data base using mobile phone tracking information provided by the three main phone operators in Spain with the aim of tracking population movements in the context of the pandemic.

This mobility data base contains daily information on the share of the mobile population at the province and CC.AA level during lockdown between March 16th and June 2020th[31], and during a reference period - the week of the 18th of November 2019. For each considered geographical level, the mobile population is defined as the population which left its residential area for more than 2 hours between 10am and 4pm. Residential areas were defined by the INE based on census delimitations and comprise between 5,000 and 50,000 inhabitants. The largest city in Spain, Madrid, is divided into 150 residential areas of approximately 20,000 inhabitants.

Mobility reduction indicators

We first construct an indicator of mobility reduction at the province and monthly level, which is equal to the difference between the share of the mobile population in province p and month m and the share of the mobile population in province p in November 2019. We denote it $MobRed_{p,m}$.

For each province, we then proceed to average out $MobRed_{p,m}$ and take the absolute value to obtain a measure of the stringency of the confinement at the province level between March and

[31] Daily mobility data in March is given for every other day, and for everyday in the following months.

June 2020. We denote it $String_p$. The higher the value of $String_p$, the lower mobility was during confinement compared to pre-pandemic levels.

We then compute the median value of $String_p$ across all provinces and classify provinces into high and low mobility reduction according to their respective value of $String_p$. Provinces with the highest stringency indicator (above median) are classified into high mobility reduction provinces, and vice versa.

The determinants of effective mobility across provinces

Mobility reduction differences during confinement could steam from pre-pandemic province characteristics (such as demographics, labor market structure, etc.) or from differences in the evolution of the pandemic across the Spanish territory– with mobility being more restricted in the most hit provinces, either because of a stronger enforcement of the confinement measures by the local police or because of a stricter quarantine compliance by the population due to the fear of being infected/spreading the virus. We formally quantify to what extent mobility reduction reflects other area characteristics by running an OLS regression of the monthly decrease in mobility compared to November 2019 ($MobRed_{p,m}$) on yearly population density, quarterly unemployment rates, and COVID-19 incidence as measured by the monthly number of new-cases per 100,000 inhabitants. We note that COVID-19 incidence is endogenous to mobility reduction, as mobility reduction affects the diffusion of the virus and as higher incidence of COVID-19 cases is likely to encourage people to stay at home (or policy makers to strengthen social distancing measures). Nonetheless, results displayed in the table suggest that none of the explanatory variables considered explain significant differences between provinces, once province and time fixed effects are controlled for (specification 6).

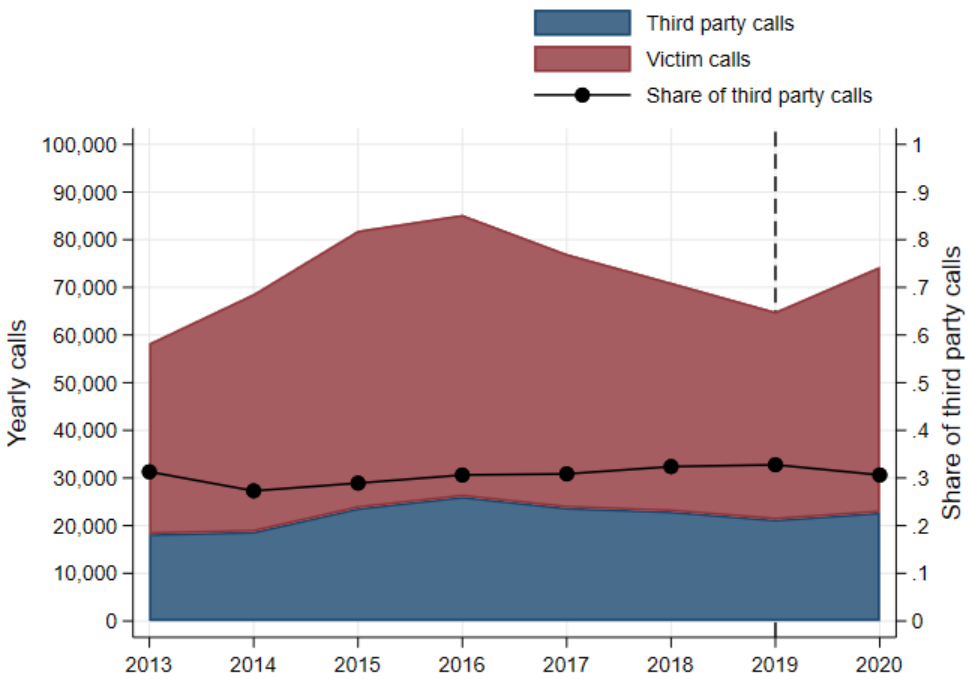
Table A 1. Factors of variation of mobility reduction across provinces between March and June 2020: OLS estimates

	(1) Mobility reduction, density	(2) Mobility reduction, COVID-19 incidence	(3) Mobility reduction, unemployment	(4) Mobility reduction, all controls w/o FE	(5) Mobility reduction, all controls w/ province FE	(6) Mobility reduction, all controls and FE
Population density	0.081			-0.001***	0.289	0.089
	(0.085)			(0.000)	(0.375)	(0.089)
Monthly COVID-19 cases per 100,000		0.001		-0.015***	-0.017***	0.001
		(0.001)		(0.003)	(0.003)	(0.001)
Quaterly Female unemployment rate			-0.073	0.450***	0.148	-0.105
			(0.101)	(0.118)	(0.409)	(0.087)
Quaterly Male unemployment rate			0.109	-0.598**	-1.243**	0.068
			(0.119)	(0.180)	(0.433)	(0.114)
Constant	-27.114	0.000	-0.138	-9.404***	-2.131	-28.915
	(28.578)	(0.386)	(1.993)	(1.239)	(11.309)	(29.496)
Observations	260	260	260	260	260	260
Within R^2	0.953	0.953	0.953	0.259	0.400	0.954
Province FE	Yes	Yes	Yes	No	Yes	Yes
Month and year FE	Yes	Yes	Yes	No	No	Yes

Note: The outcome of interest is the monthly decrease in the share of the mobile population in province p and month m, compared to the share of the mobile population in November 2019 in province p. We regress the monthly decrease in mobility compared to November 2019 on population density, COVID-19 incidence as measured by the monthly number of cases per 100,000 inhabitants, quarterly male and female unemployment rate. *Standard errors are clustered at the province level.*
Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Tables and figures

Fig A 1. Yearly number of calls according to type of caller over all Spanish provinces, 2013-2020



Source: National Office for gender-based violence

Fig A 2. Yearly number of calls according to type of caller over all Spanish provinces, 2013-2020

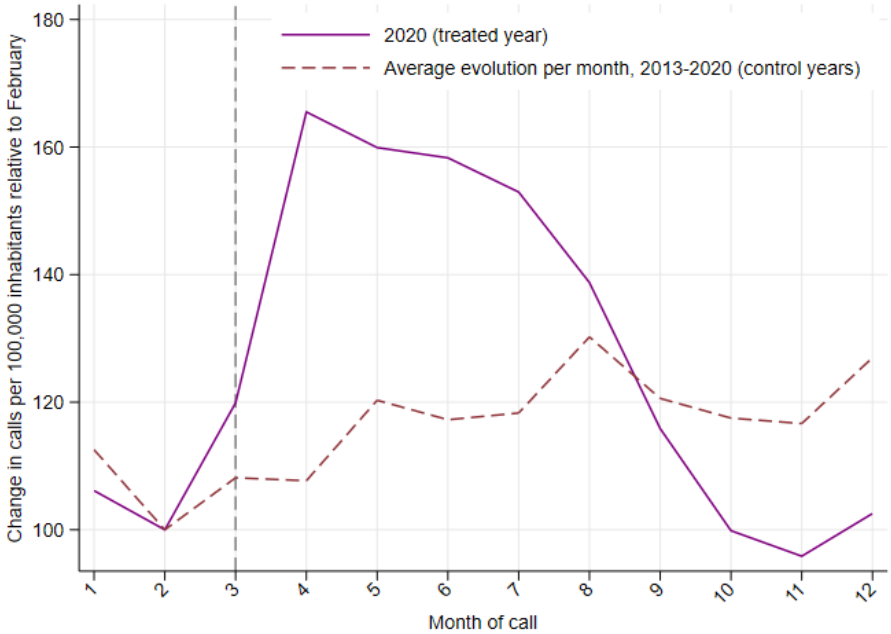


Table A 2. Total calls and calls per provinces between March and December, on average over the control years (2013-2019)

	Average yearly calls in Spain		**Average yearly calls per provinces,**	
Caller	*Number of calls*	*Calls per 100 000 inhabitants*	*Number of calls*	*Calls per 100 000 inhabitants*
Total	55230	118.7	1062.1	100.1
Victim	39480	84.9	759.2	71.1
Third party	15750	33.9	302.8	29.0

Table A 3. Impact of lockdown introduction on <u>total</u> domestic violence reporting through emergency hotline – difference-in-differences estimates: log models with controls

	(1) Log of calls, controlling for density	(2) Log of calls, controlling for unemp rate	(3) Log of calls, controlling for male and female unemp rate	(4) Log of calls, all controls
$\boldsymbol{D_{y=2020}D_{m\geq 3}}$	**0.390*** **	**0.389*** **	**0.388*** **	**0.389*** **
	(0.026)	**(0.026)**	**(0.027)**	**(0.027)**
$\boldsymbol{D_{m\geq 3}}$	0.040+	0.037+	0.038+	0.039+
	(0.023)	(0.021)	(0.022)	(0.021)
$\boldsymbol{D_{y=2020}}$	0.035	0.017	0.021	0.027
	(0.025)	(0.038)	(0.040)	(0.041)
Population density in 2019	-0.001***			-0.001*
	(0.000)			(0.000)
Quarterly unemp. rate		-0.002		
		(0.003)		
Quarterly female unemp. rate			-0.002	-0.001
			(0.002)	(0.003)
Quaterly male unemp. rate			0.000	0.000
			(0.003)	(0.003)
Constant	3.708***	3.470***	3.464***	3.699***
	(0.330)	(0.309)	(0.313)	(0.336)
Observations	4990	4990	4990	4990
R^2	0.216	0.215	0.215	0.216
Province FE	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes

<u>Note</u>: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March, and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years. Standard errors are clustered at the province level.
<u>Significance level</u>: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table A 4. Impact of lockdown introduction on victims' domestic violence reporting through emergency hotline – difference-in-differences estimates: log models with controls

	(1) Log of calls, controlling for density	(2) Log of calls, controlling for unemp rate	(3) Log of calls, controlling for male and female unemp.	(4) Log of calls, all controls
$\boldsymbol{D_{y=2020} D_{m\geq 3}}$	**0.380*** **	**0.381*** **	**0.377*** **	**0.378*** **
	(0.030)	**(0.030)**	**(0.031)**	**(0.031)**
$\boldsymbol{D_{m\geq 3}}$	0.035	0.038+	0.041+	0.042*
	(0.023)	(0.021)	(0.021)	(0.021)
$\boldsymbol{D_{y=2020}}$	0.062*	0.084+	0.095+	0.103*
	(0.030)	(0.045)	(0.047)	(0.048)
Population density in 2019	-0.002***			-0.002*
	(0.000)			(0.001)
Quarterly unemp. rate		0.002		
		(0.003)		
Quarterly female unemp. rate			-0.003	-0.002
			(0.002)	(0.003)
Quaterly male unemp. rate			0.005	0.005
			(0.003)	(0.003)
Constant	3.406***	2.999***	2.983***	3.313***
	(0.339)	(0.308)	(0.320)	(0.355)
Observations	4984	4984	4984	4984
R^2	0.177	0.177	0.178	0.178
Province FE	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March, and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years. Standard errors are clustered at the province level.
Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table A 5. Impact of lockdown introduction on third parties' domestic violence reporting through emergency hotline – difference-in-differences estimates: log models with controls

	(1) Log of calls, controlling for density	(2) Log of calls, controlling for unemp rate	(3) Log of calls, controlling for male and female unemp	(4) Log of calls, all controls
$\boldsymbol{D_{y=2020} D_{m\geq 3}}$	**0.414*** **	**0.381*** **	**0.377*** **	**0.378*** **
	(0.035)	**(0.030)**	**(0.031)**	**(0.031)**
$\boldsymbol{D_{m\geq 3}}$	0.028	0.038+	0.041+	0.042*
	(0.029)	(0.021)	(0.021)	(0.021)
$\boldsymbol{D_{y=2020}}$	-0.029	0.084+	0.095+	0.103*
	(0.029)	(0.045)	(0.047)	(0.048)
Population density in 2019	-0.002***			-0.002*
	(0.000)			(0.001)
Quarterly unemp. rate		0.002		
		(0.003)		
Quarterly female unemp. rate			-0.003	-0.002
			(0.002)	(0.003)
Quaterly male unemp. rate			0.005	0.005
			(0.003)	(0.003)
Constant	2.608***	2.999***	2.983***	3.313***
	(0.387)	(0.308)	(0.320)	(0.355)
Observations	4949	4984	4984	4984
R^2	0.133	0.177	0.178	0.178
Province FE	Yes	Yes	Yes	Yes
Month and year FE	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{m\geq 3}$ is a dummy variable which is equal to 0 for the months of January and February, 0.5 for March, and 1 from April to December. β_{did} is our coefficient of interest: it captures the effect of lockdown introduction on the 15th of March 2020 onwards compared to previous years. Standard errors are clustered at the province level.
Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Fig A 3. Impact of lockdown introduction on <u>total</u> domestic violence reporting through emergency hotline – event study estimates: log model

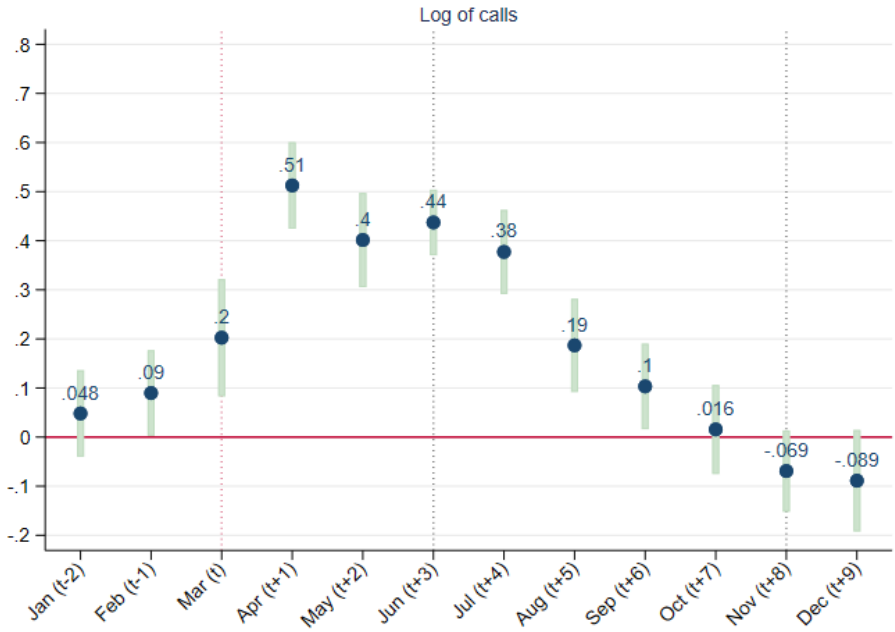


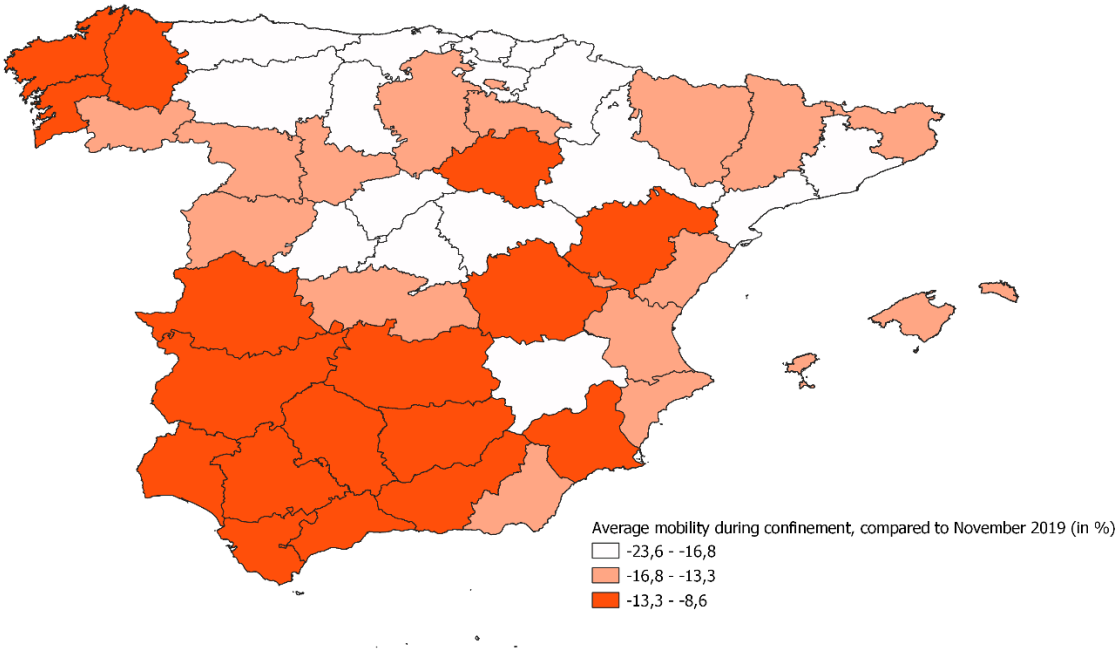
Table A 6. Share of the mobile population and mobility reduction in Spain, per month

Month	Average share of the mobile population across provinces	Average decrease of the mobile population across provinces
November (reference month) 2019	26.9%	0.0pp
March 2020	9.5%	-17.4pp
April 2020	8.6%	-18.3pp
May 2020	12.6%	-14.3pp
June 2020	16.6%	-10.3pp

Source: INE, Statistical information for the analysis of the COVID-19 crisis' impact. Mobility data.

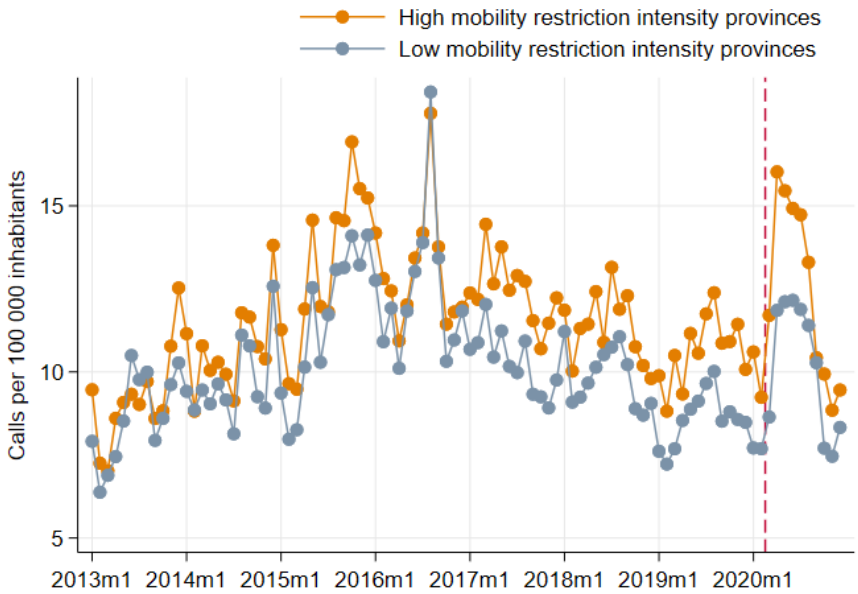
Note: Mobility reduction is the difference between the share of the mobile population in the considered month, with the share of the mobile population in the reference month (November)

Fig A 4. Average decrease in the share of the mobile population between the 15th of March 2020 and the 20th of June, compared to November 2019



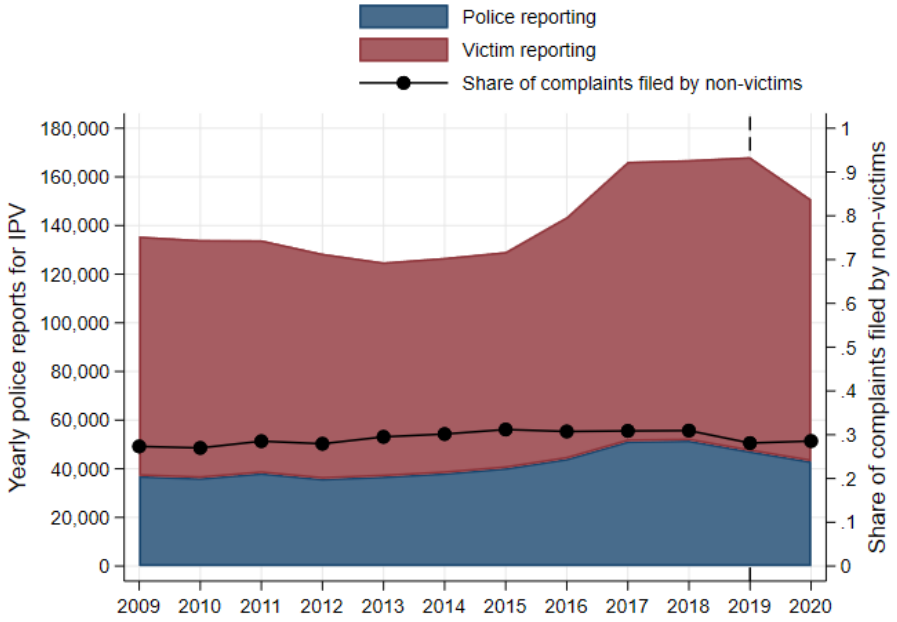
Source: INE

Fig A 5. Monthly calls per 100,000 inhabitants between 2013 and 2020, in high and low mobility reduction provinces



Source: INE

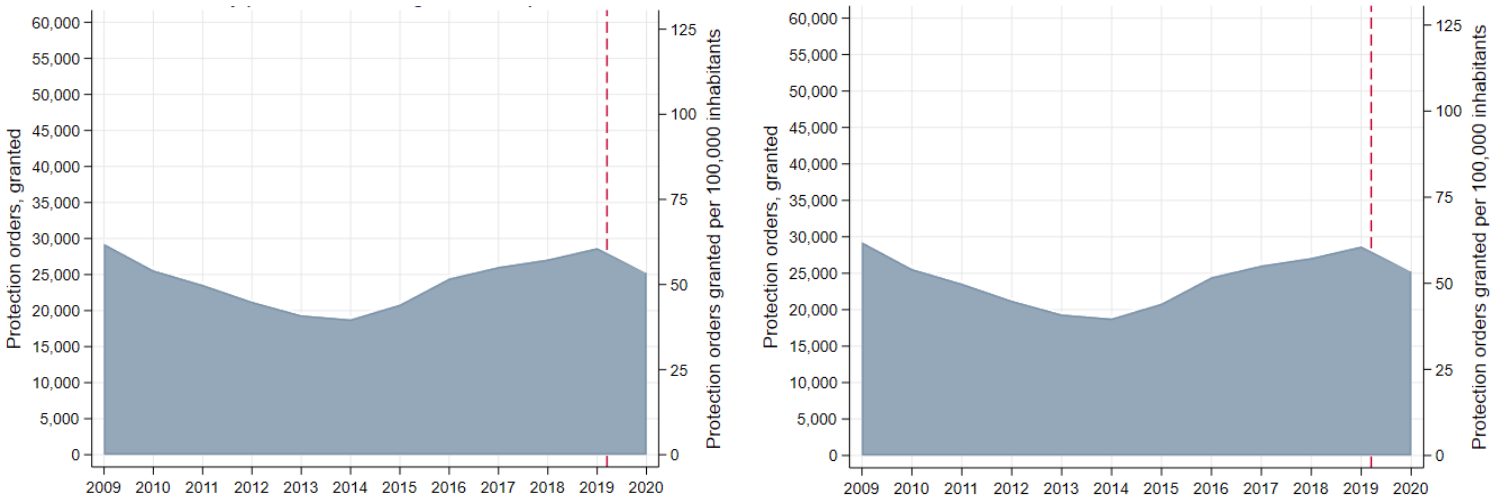
Fig A 6 Yearly number of police complaints according to who filed the complaint between 2009 and 2020



Source: Poder Judicial (CGPJ)

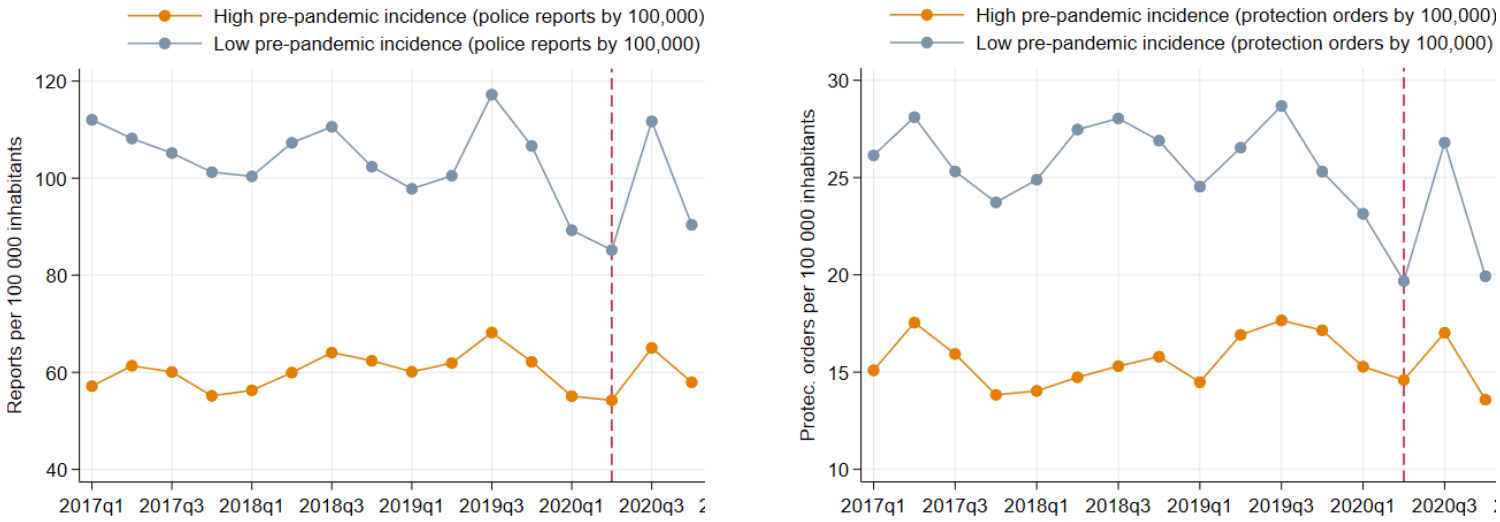
Fig A 7. Yearly number of protection orders and protection orders per 100,000 inhabitants between 2009 and 2020, applications and orders granted

Applications for protection orders ***Protection orders granted by the judge***



Source: Poder Judicial (CGPJ)

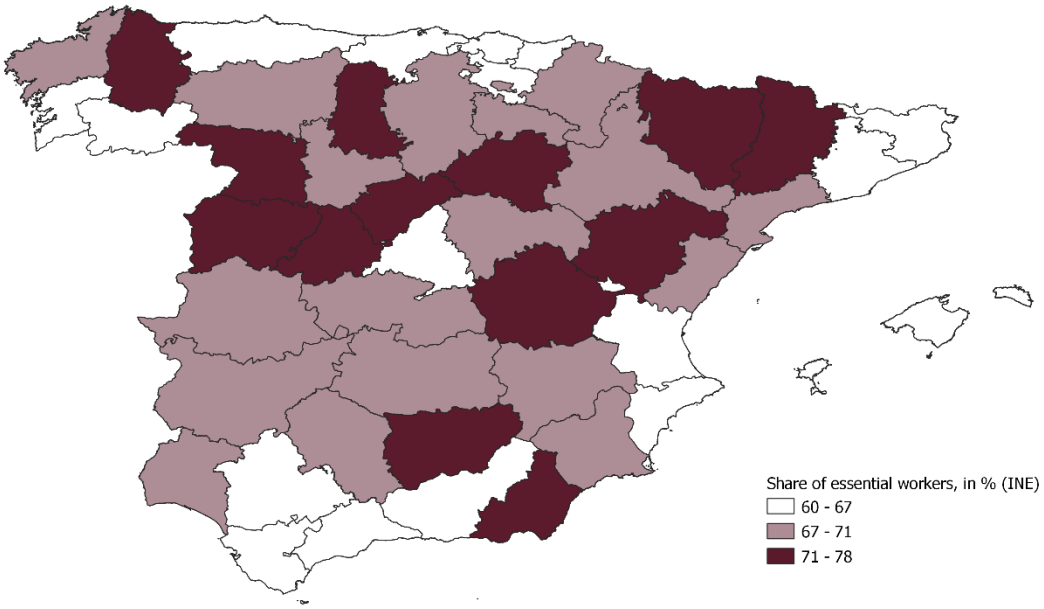
Fig A 8. Pre-pandemic IPV incidence as measured by police reports and protection orders per 100,000 inhabitants



Source: INE

Note: for each considered measure of IPV incidence (police reports or protection orders), high vs low pre-pandemic IPV incidence categories are defined based on the average value of IPV incidence per 100,000 inhabitants between 2015 and 2018. Provinces with average values above (resp. below or equal to) the median are considered as high (resp. low) pre-pandemic IPV incidence provinces.

Fig A 9. Share of essential workers by province, as per Royal Decree 10/2020 (INE)



Source: INE

Table A 7. Effect of lockdown introduction on quarterly complaints for IPV and protection orders, DiD estimates

	(1) Complaints (log)	(2) Complaints filed by victims (log)	(3) Applications for protec. Orders (log)	(4) Orders granted (log)	(5) Complaints per 100,000	(6) Complaints *filed by victims* per 100,000	(7) Applications for orders per 100,000	(8) Orders granted per 100,000
$\boldsymbol{D_{y=2020}D_{q\geq 2}}$	0.015	0.037	-0.114**	-0.096*	1.913	2.455+	-1.710**	-1.127+
	(0.025)	(0.034)	(0.033)	(0.037)	(1.506)	(1.391)	(0.626)	(0.588)
$\boldsymbol{D_{q\geq 2}}$	0.001	0.004	0.007	-0.004	0.285	-0.147	-0.030	-0.222
	(0.009)	(0.016)	(0.013)	(0.012)	(0.628)	(0.625)	(0.210)	(0.177)
$\boldsymbol{D_{y=2020}}$	0.076+	0.060	0.027	-0.001	3.239	1.794	-0.162	-0.186
	(0.039)	(0.050)	(0.072)	(0.092)	(2.802)	(2.430)	(1.386)	(1.329)
Constant	5.827***	5.725***	4.814***	4.537***	65.004***	49.237***	18.918***	14.770***
	(0.171)	(0.214)	(0.352)	(0.426)	(1.775)	(1.493)	(0.749)	(0.715)
Mean value (*)					*69.40*	*50.37*	*19.24*	*13.79*
Observations	2496	2496	2496	2496	2496	2496	2496	2496
R^2	0.301	0.132	0.091	0.125	0.288	0.204	0.099	0.130
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter and year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: $D_{y=2020}$ is a dummy variable which is equal to 1 for the year 2020, and 0 otherwise; $D_{q\geq 2}$ is a dummy variable which is equal to 0 from January to March, and 1 for the 2nd 3rd and 4th quarter of year. $\boldsymbol{D_{y=2020}D_{q\geq 2}}$ *is our coefficient of interest: it captures the effect of lockdown introduction from the 2nd quarter of year onwards compared to previous years. Standard errors are clustered at the province level.*
(*) The mean value displayed corresponds to the quarterly value of the outcome per 100,000 inhabitants in the pre-treatment period (for Q=2,3 or 4), on average in the control period (2013-2019).
Significance level: + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

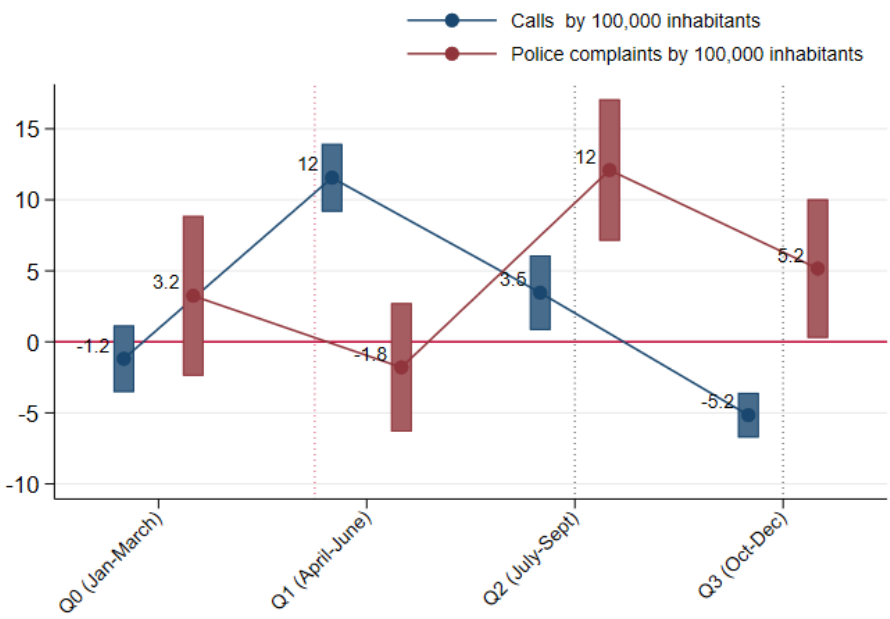



Fig A 10. Impact of lockdown introduction on total complaints and calls to the hotline per 100,000 inhabitants– event study estimates